

Atomic Structure

Rutherford's Nuclear Model of the Atom

Postulates

1. Every atom contains a small, dense and positively charged nucleus at its centre.
2. Nearly all the mass of the atom is concentrated in the nucleus.
3. Electrons revolve around the nucleus in circular paths.
4. The nucleus occupies only a very small volume compared to the size of the atom.
5. Most of the atom consists of empty space.
6. The atom as a whole is electrically neutral because the total positive charge is equal to the total negative charge.

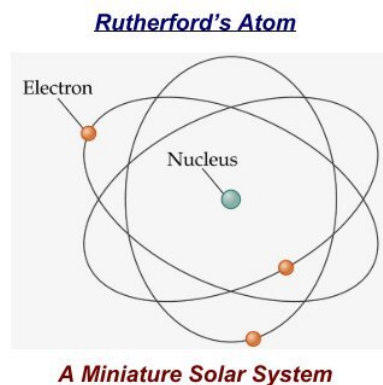


Figure 1: Rutherford's Solar System Model of the Atom

1 Atomic Number and Mass Number

Atomic Number

The atomic number of an element is the number of protons present in the nucleus of an atom.

It is denoted by Z .

$$\text{Atomic Number} = \text{Number of Protons}$$

For a neutral atom,

$$\text{Number of Electrons} = \text{Number of Protons}$$

Mass Number

The mass number of an atom is the total number of protons and neutrons present in the nucleus.

It is denoted by A .

$$\text{Mass Number} = \text{Number of Protons} + \text{Number of Neutrons}$$

or

$$A = Z + N$$

where

- A = Mass Number
- Z = Atomic Number
- N = Number of Neutrons

Therefore,

$$N = A - Z$$

Worked Example 1

Find the number of protons, neutrons and electrons in ${}_{11}^{23}\text{Na}$.

Solution

$$Z = 11$$

$$A = 23$$

Number of protons

$$= Z = 11$$

Number of electrons

$$= 11$$

Number of neutrons

$$= A - Z$$

$$= 23 - 11 = 12$$

Answer:

- Protons = 11
- Electrons = 11
- Neutrons = 12

Worked Example 2

Find the number of protons, neutrons and electrons in ${}^{35}_{17}\text{Cl}^-$.

Solution

$$Z = 17$$

$$A = 35$$

Number of protons

$$= 17$$

Number of neutrons

$$= 35 - 17 = 18$$

Since chloride ion has one extra electron,

$$\text{Electrons} = 17 + 1 = 18$$

Answer:

- Protons = 17
- Electrons = 18
- Neutrons = 18

Practice Examples

1. ${}^{16}_8\text{O}$

- Protons = 8
- Electrons = 8
- Neutrons = $16 - 8 = 8$

2. ${}^{24}_{12}\text{Mg}$

- Protons = 12
- Electrons = 12
- Neutrons = $24 - 12 = 12$

3. ${}^{40}_{20}\text{Ca}^{2+}$

- Protons = 20
- Electrons = $20 - 2 = 18$
- Neutrons = $40 - 20 = 20$

4. ${}^{27}_{13}\text{Al}^{3+}$

- Protons = 13
- Electrons = $13 - 3 = 10$
- Neutrons = $27 - 13 = 14$

5. ${}^{19}_{9}\text{F}^{-}$

- Protons = 9
- Electrons = $9 + 1 = 10$
- Neutrons = $19 - 9 = 10$

6. ${}^{32}_{16}\text{S}^{2-}$

- Protons = 16
- Electrons = $16 + 2 = 18$
- Neutrons = $32 - 16 = 16$

7. ${}^{39}_{19}\text{K}^{+}$

- Protons = 19
- Electrons = $19 - 1 = 18$
- Neutrons = $39 - 19 = 20$

Important Points

- Atomic Number (Z) = Number of Protons.
- Mass Number (A) = Number of Protons + Number of Neutrons.
- For neutral atoms, Number of Electrons = Number of Protons.
- For positive ions (cations), electrons are lost.
- For negative ions (anions), electrons are gained.

2 Isotopes and Isobars

Isotopes

Atoms of the same element having the same atomic number but different mass numbers are called **isotopes**.

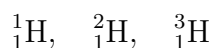
Characteristics of Isotopes

- They have the same atomic number.
- They have the same number of protons and electrons.
- They have different numbers of neutrons.
- They have different mass numbers.

- They exhibit similar chemical properties.

Examples

1. Hydrogen isotopes:



These are called Protium, Deuterium and Tritium.

2. Chlorine isotopes:

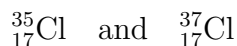


3. Carbon isotopes:



Worked Example

Consider the isotopes



- Atomic number = 17 for both atoms.
- Number of protons = 17 for both atoms.
- Number of neutrons in ${}^{35}_{17}\text{Cl} = 35 - 17 = 18$.
- Number of neutrons in ${}^{37}_{17}\text{Cl} = 37 - 17 = 20$.

Hence they are isotopes.

Isobars

Atoms of different elements having the same mass number but different atomic numbers are called **isobars**.

Characteristics of Isobars

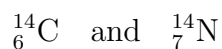
- They have different atomic numbers.
- They have different numbers of protons.
- They belong to different elements.
- They have the same mass number.
- They exhibit different chemical properties.

Examples

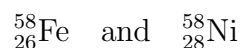
1.



2.



3.



Worked Example

Consider



- Mass number = 40 for both atoms.
- Atomic numbers are different (18 and 20).
- Therefore, they are different elements.

Hence they are isobars.

Differences Between Isotopes and Isobars

Property	Isotopes	Isobars
Atomic Number	Same	Different
Mass Number	Different	Same
Element	Same	Different
Chemical Properties	Similar	Different

Important Points

- Isotopes have the same atomic number but different mass numbers.
- Isobars have the same mass number but different atomic numbers.
- Isotopes belong to the same element.
- Isobars belong to different elements.

3 Electromagnetic Spectrum

Generation of Electromagnetic Radiation

Electromagnetic radiation is produced when an electrically charged particle, such as an electron, is accelerated.

When electrons gain energy, they move to higher energy levels. As they return to lower energy levels, the excess energy is emitted in the form of electromagnetic radiation.

Electromagnetic radiation does not require a material medium for its propagation and can travel through vacuum with the speed of light.

Electromagnetic Spectrum

The arrangement of different types of electromagnetic radiations in the order of increasing wavelength or decreasing frequency is called the **electromagnetic spectrum**.

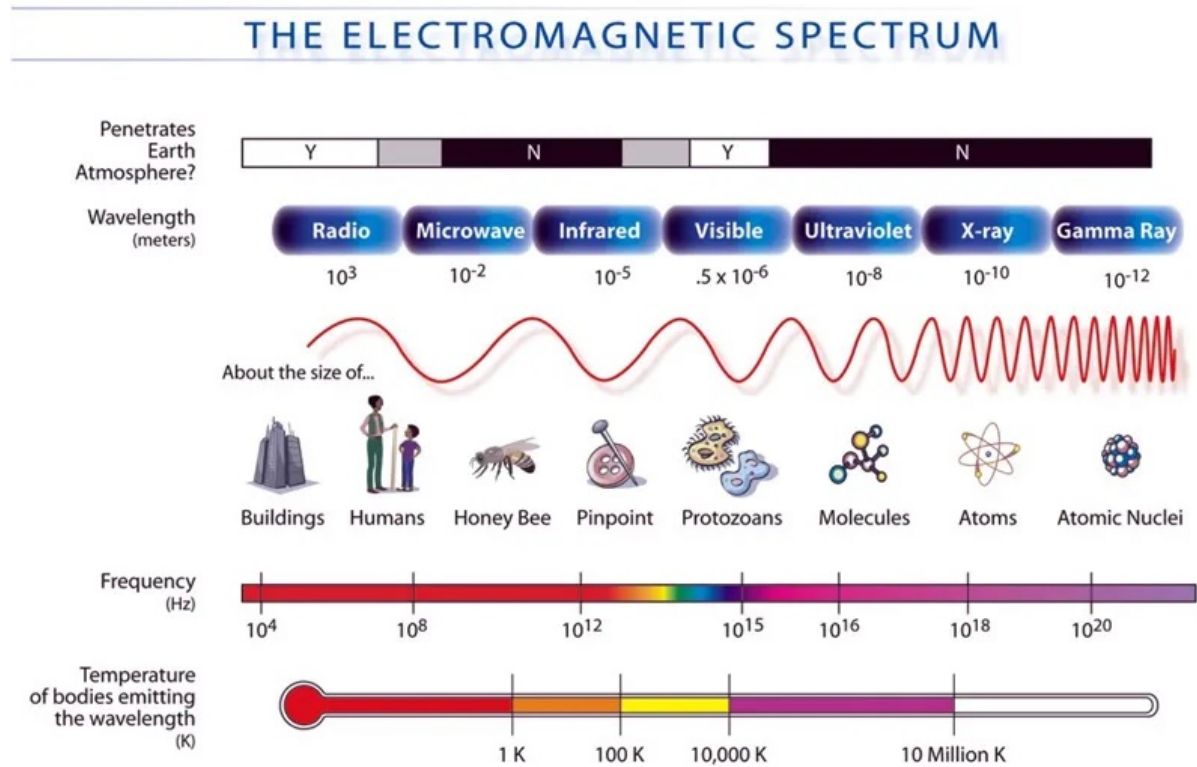


Figure 2: Electromagnetic Spectrum

4 Relationship Between Energy, Frequency and Wavelength

The energy of electromagnetic radiation depends on its frequency and wavelength.

Important Equations

Planck's Equation

$$E = h\nu$$

where

- E = Energy of radiation (J)
- h = Planck's constant = 6.626×10^{-34} Js
- ν = Frequency (s^{-1})

Relationship Between Frequency and Wavelength

$$c = \nu\lambda$$

where

- c = Speed of light = $3.0 \times 10^8 \text{ ms}^{-1}$
- ν = Frequency
- λ = Wavelength

Combining the above equations,

$$E = \frac{hc}{\lambda}$$

Thus,

- Energy is directly proportional to frequency.
- Energy is inversely proportional to wavelength.

Solved Example 1

Problem:

Calculate the energy of radiation having a frequency of

$$6.0 \times 10^{14} \text{ s}^{-1}$$

Solution:

Using

$$E = h\nu$$

$$E = (6.626 \times 10^{-34})(6.0 \times 10^{14})$$

$$E = 3.98 \times 10^{-19} \text{ J}$$

Answer:

$$E = 3.98 \times 10^{-19} \text{ J}$$

Solved Example 2

Problem:

Calculate the frequency of radiation whose wavelength is

$$500 \text{ nm}$$

Solution:

Convert wavelength into metres:

$$500 \text{ nm} = 500 \times 10^{-9} \text{ m}$$

Using

$$c = \nu \lambda$$

$$\nu = \frac{c}{\lambda}$$

$$\nu = \frac{3.0 \times 10^8}{500 \times 10^{-9}}$$

$$\nu = 6.0 \times 10^{14} \text{ s}^{-1}$$

Answer:

$$\nu = 6.0 \times 10^{14} \text{ s}^{-1}$$

Solved Example 3

Problem:

Calculate the energy of radiation having a wavelength of

$$500 \text{ nm}$$

Solution:

Using

$$E = \frac{hc}{\lambda}$$

$$E = \frac{(6.626 \times 10^{-34})(3.0 \times 10^8)}{500 \times 10^{-9}}$$

$$E = 3.98 \times 10^{-19} \text{ J}$$

Answer:

$$E = 3.98 \times 10^{-19} \text{ J}$$

Quick Revision

$$E = h\nu$$

$$c = \nu\lambda$$

$$E = \frac{hc}{\lambda}$$

- Higher frequency $\rightarrow$ Higher energy
- Longer wavelength $\rightarrow$ Lower energy
- Shorter wavelength $\rightarrow$ Higher energy

5 Bohr's Atomic Model

Postulates

1. Electrons revolve around the nucleus only in certain permitted circular orbits called stationary orbits.
2. Each stationary orbit is associated with a definite amount of energy.
3. While revolving in a stationary orbit, an electron neither emits nor absorbs energy.
4. The energy of an electron increases as its distance from the nucleus increases.
5. Energy is absorbed when an electron jumps from a lower energy orbit to a higher energy orbit.
6. Energy is emitted when an electron jumps from a higher energy orbit to a lower energy orbit.
7. The energy absorbed or emitted during a transition is given by

$$\Delta E = E_2 - E_1 = h\nu$$

where

- h = Planck's constant
- ν = Frequency of radiation

8. The angular momentum of an electron in a stationary orbit is quantized and is given by

$$mvr = \frac{nh}{2\pi}$$

where

- m = Mass of the electron
- v = Velocity of the electron
- r = Radius of the orbit
- n = Principal quantum number
- h = Planck's constant

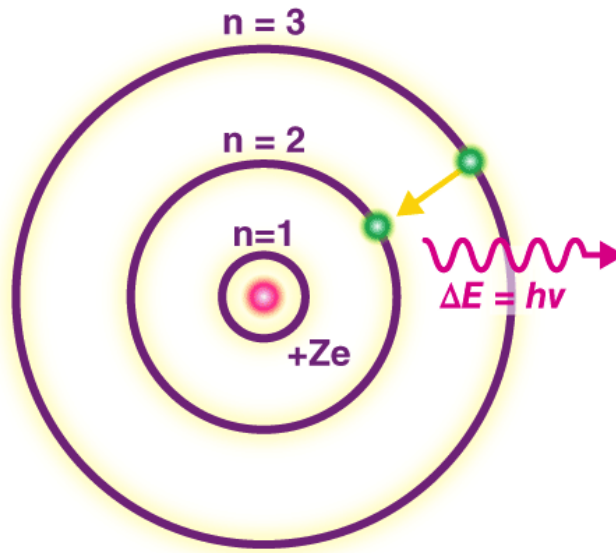


Figure 3: Bohr's Atomic Model

6 Absorption Spectrum and Emission Spectrum

Absorption Spectrum

When white light passes through a cool gaseous substance, certain wavelengths are absorbed by the atoms of the gas. As a result, dark lines are observed on a bright background. This spectrum is called an **absorption spectrum**.

- Produced when atoms absorb energy.
- Consists of dark lines on a bright background.
- The absorbed wavelengths are missing from the spectrum.

Emission Spectrum

When an excited gaseous substance is allowed to cool, its atoms emit radiation of specific wavelengths. This produces bright lines on a dark background. This spectrum is called an **emission spectrum**.

- Produced when atoms emit energy.
- Consists of bright lines on a dark background.
- The emitted wavelengths appear as bright lines.

Differences Between Absorption and Emission Spectra

Property	Absorption Spectrum	Emission Spectrum
Formation	Produced when atoms absorb radiation	Produced when atoms emit radiation
Appearance	Dark lines on a bright background	Bright lines on a dark background
Energy Change	Electrons move to higher energy levels	Electrons move to lower energy levels
Type of Lines	Dark lines	Bright lines

Important Points

- Absorption and emission spectra are line spectra.
- Each element has a characteristic spectrum.
- Spectra can be used to identify elements.

7 de Broglie's Wave-Particle Duality

Light exhibits both wave nature and particle nature. In 1924, de Broglie extended this concept to all moving particles and proposed that every moving particle is associated with a wave called a matter wave.

de Broglie Equation

The wavelength associated with a moving particle is given by

$$\lambda = \frac{h}{mv}$$

or

$$\lambda = \frac{h}{p}$$

where

- λ = Wavelength of the particle
- h = Planck's constant
- m = Mass of the particle
- v = Velocity of the particle
- p = Momentum of the particle

Important Points

- Every moving particle possesses wave nature.
- These waves are called matter waves.
- The wavelength is inversely proportional to the momentum of the particle.
- Faster particles have shorter wavelengths.
- Wave nature is significant only for microscopic particles such as electrons.

Worked Example

Calculate the de Broglie wavelength of an electron moving with momentum $2.0 \times 10^{-24} \text{ kg m s}^{-1}$.

$$\begin{aligned}\lambda &= \frac{h}{p} \\ &= \frac{6.626 \times 10^{-34}}{2.0 \times 10^{-24}} \\ &= 3.31 \times 10^{-10} \text{ m}\end{aligned}$$

Answer:

$$\lambda = 3.31 \times 10^{-10} \text{ m}$$

8 Heisenberg's Uncertainty Principle

In 1927, Werner Heisenberg proposed the uncertainty principle.

Statement

It is impossible to determine simultaneously the exact position and exact momentum of a microscopic particle.

Mathematical Expression

$$\Delta x \Delta p \geq \frac{h}{4\pi}$$

where

- Δx = Uncertainty in position
- Δp = Uncertainty in momentum
- h = Planck's constant

Important Points

- The exact path of an electron cannot be determined.
- Greater accuracy in position results in greater uncertainty in momentum.
- Greater accuracy in momentum results in greater uncertainty in position.
- The principle is important for microscopic particles.
- It has negligible effect on macroscopic objects.

9 Photoelectric Effect

Definition

The emission of electrons from a metal surface when light of suitable frequency falls on it is called the photoelectric effect.

Condition for Photoelectric Effect

The frequency of incident radiation must be greater than or equal to the threshold frequency of the metal.

$$\nu \geq \nu_0$$

where

- ν = Frequency of incident radiation
- ν_0 = Threshold frequency

Einstein's Photoelectric Equation

$$h\nu = h\nu_0 + \frac{1}{2}mv^2$$

or

$$h\nu = \phi + KE$$

where

- $h\nu$ = Energy of incident photon
- ϕ = Work function
- KE = Maximum kinetic energy of the emitted electron

Important Points

- Einstein explained the photoelectric effect using photon theory.
- The number of emitted electrons depends on the intensity of light.
- The kinetic energy of photoelectrons depends on the frequency of light.
- No photoelectric emission occurs below the threshold frequency.

10 Quantum Numbers

According to the quantum mechanical model, the complete description of an electron in an atom requires four quantum numbers. These quantum numbers specify the energy, shape, orientation, and spin of an electron.

The four quantum numbers are:

1. Principal Quantum Number (n)
2. Azimuthal Quantum Number (l)
3. Magnetic Quantum Number (m_l)
4. Spin Quantum Number (m_s)

1. Principal Quantum Number (n)

The principal quantum number specifies the main energy level (shell) occupied by an electron.

Important Points

- It is represented by the symbol n .
- It determines the shell in which the electron is present.
- It determines the size and energy of the orbital.
- As the value of n increases, the size and energy of the orbital also increase.
- The maximum number of electrons present in a shell is given by

$$2n^2$$

Possible Values

$$n = 1, 2, 3, 4, \dots$$

n	Shell	Maximum Electrons
1	K	2
2	L	8
3	M	18
4	N	32
5	O	50

Examples

1. Electron in 2p orbital

$$n = 2$$

2. Electron in 4d orbital

$$n = 4$$

3. Electron in 5f orbital

$$n = 5$$

2. Azimuthal Quantum Number (l)

The azimuthal quantum number specifies the subshell and the shape of the orbital.

Important Points

- It is represented by the symbol l .
- It determines the shape of an orbital.
- It specifies the subshell (s, p, d or f).
- The possible values depend on the value of the principal quantum number.
- The number of subshells in a shell is equal to the value of n .

Possible Values

$$l = 0, 1, 2, \dots, (n - 1)$$

l	Subshell	Shape
0	s	Spherical
1	p	Dumbbell
2	d	Cloverleaf
3	f	Complex

Examples

1. Electron in 3s orbital

$$l = 0$$

2. Electron in 4p orbital

$$l = 1$$

3. Electron in 5d orbital

$$l = 2$$

3. Magnetic Quantum Number (m_l)

The magnetic quantum number specifies the orientation of an orbital in space.

Important Points

- It is represented by the symbol m_l .
- It determines the orientation of an orbital.
- Its values depend upon the azimuthal quantum number.
- The total number of orbitals in a subshell is given by

$$2l + 1$$

Possible Values

$$m_l = -l, \dots, 0, \dots, +l$$

Examples:

$$l = 0 \Rightarrow m_l = 0$$

$$l = 1 \Rightarrow m_l = -1, 0, +1$$

$$l = 2 \Rightarrow m_l = -2, -1, 0, +1, +2$$

$$l = 3 \Rightarrow m_l = -3, -2, -1, 0, +1, +2, +3$$

Subshell	Number of Orbitals
s	1
p	3
d	5
f	7

Examples

1. s-subshell

$$m_l = 0$$

2. p-subshell

$$m_l = -1, 0, +1$$

3. d-subshell

$$m_l = -2, -1, 0, +1, +2$$

4. Spin Quantum Number (m_s)

The spin quantum number specifies the direction of spin of an electron.

Important Points

- It is represented by the symbol m_s .
- It specifies the direction of electron spin.
- An electron may spin in one of two possible directions.
- Two electrons in the same orbital always have opposite spins (Pauli Exclusion Principle).

Possible Values

$$m_s = +\frac{1}{2}$$

or

$$m_s = -\frac{1}{2}$$

Examples

1. First electron in an orbital

$$m_s = +\frac{1}{2}$$

2. Second electron in the same orbital

$$m_s = -\frac{1}{2}$$

3. Any electron can have only one of these two values.

Summary Table

Quantum Number	Represents	Possible Values
Principal (n)	Shell / Energy level	1, 2, 3, ...
Azimuthal (l)	Subshell / Shape	0 to ($n - 1$)
Magnetic (m_l)	Orientation of orbital	$-l$ to $+l$
Spin (m_s)	Spin of electron	$\pm\frac{1}{2}$

Solved Examples

Example 1

Find the quantum numbers of an electron present in the 2s orbital.

Solution

For a 2s orbital,

$$n = 2$$

Since it is an s-orbital,

$$l = 0$$

For an s-orbital,

$$m_l = 0$$

The spin can be

$$m_s = +\frac{1}{2}$$

or

$$m_s = -\frac{1}{2}$$

Answer

$$n = 2, l = 0, m_l = 0, m_s = \pm\frac{1}{2}$$

Example 2

Find the quantum numbers of an electron present in the 3p orbital.

Solution

For a 3p orbital,

$$n = 3$$

For a p-orbital,

$$l = 1$$

Possible values of m_l are

$$-1, 0, +1$$

Possible values of m_s are

$$+\frac{1}{2}, -\frac{1}{2}$$

Answer

$$n = 3, l = 1, m_l = -1, 0, +1, m_s = \pm\frac{1}{2}$$

Example 3

An electron has the following quantum numbers:

$$n = 4, l = 2, m_l = -1, m_s = +\frac{1}{2}$$

Identify the shell and subshell.

Solution

Since

$$n = 4$$

the electron belongs to the fourth shell (N-shell).

Since

$$l = 2$$

it belongs to the d-subshell.

Answer

The electron is present in the **4d orbital**.

Example 4

An electron has

$$n = 5, l = 3, m_l = +2, m_s = -\frac{1}{2}$$

Identify the orbital.

Solution

$$n = 5$$

indicates the fifth shell.

$$l = 3$$

represents an f-subshell.

Hence,

Answer

The electron is present in the **5f orbital**.

Example 5

Determine the quantum numbers of the last electron of nitrogen ($Z = 7$).

Solution

Electronic configuration:

$$1s^2 2s^2 2p^3$$

The last electron enters the 2p subshell.

Hence,

$$n = 2$$

$$l = 1$$

According to Hund's rule, the three electrons occupy different p-orbitals.

The last electron occupies

$$m_l = +1$$

with

$$m_s = +\frac{1}{2}$$

Answer

$$n = 2, \quad l = 1, \quad m_l = +1, \quad m_s = +\frac{1}{2}$$

Practice Problems

1. Find the quantum numbers of an electron present in the 1s orbital.
2. Find the quantum numbers of an electron present in the 4p orbital.
3. Find the quantum numbers of an electron present in the 5d orbital.
4. Identify the orbital for

$$n = 3, l = 2, m_l = 0, m_s = +\frac{1}{2}$$

5. Identify the orbital for

$$n = 2, l = 1, m_l = -1, m_s = -\frac{1}{2}$$

6. Identify the orbital for

$$n = 5, l = 0, m_l = 0, m_s = +\frac{1}{2}$$

7. Determine the quantum numbers of the last electron of carbon ($Z = 6$).
8. Determine the quantum numbers of the last electron of oxygen ($Z = 8$).
9. Determine the quantum numbers of the last electron of sodium ($Z = 11$).
10. Determine the quantum numbers of the last electron of chlorine ($Z = 17$).

11 Hydrogen Emission Spectrum

When an electric discharge is passed through hydrogen gas at low pressure, its electrons absorb energy and move to higher energy levels (excited state). These excited electrons are unstable and return to lower energy levels by emitting electromagnetic radiation of definite wavelengths.

The collection of these emitted wavelengths is called the **Hydrogen Emission Spectrum**.

Rydberg Equation

The wavelength of the emitted radiation is given by

$$\boxed{\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)}$$

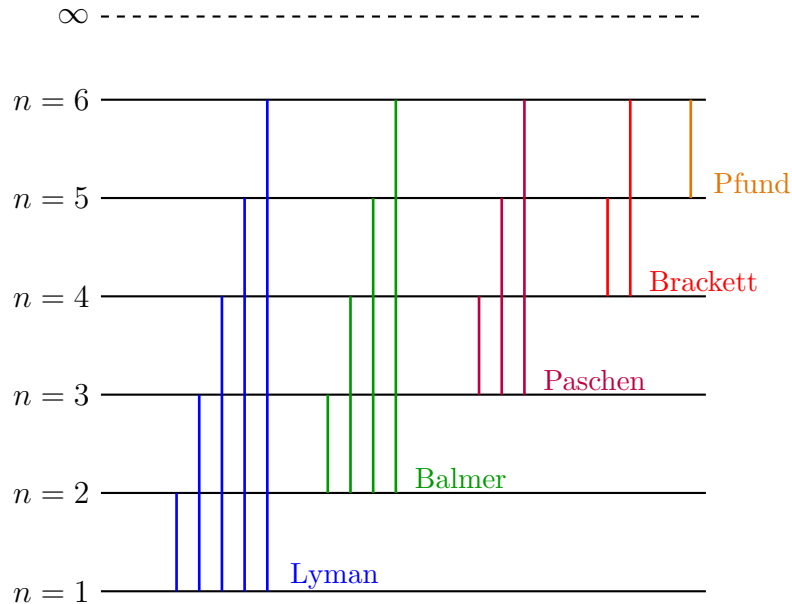
where

$$R = 1.097 \times 10^7 \text{ m}^{-1}$$

- n_i = Initial (higher) energy level

- n_f = Final (lower) energy level
- $n_i > n_f$

Hydrogen Spectral Series



Important Points

- Hydrogen emission spectrum consists of five important spectral series.
- Each spectral series corresponds to electron transitions ending at a particular energy level.
- Lyman series is produced when electrons fall to $n = 1$.
- Balmer series is produced when electrons fall to $n = 2$.
- Paschen series is produced when electrons fall to $n = 3$.
- Brackett series is produced when electrons fall to $n = 4$.
- Pfund series is produced when electrons fall to $n = 5$.
- The wavelength increases from Lyman to Pfund.
- The frequency and energy decrease from Lyman to Pfund.

Summary Table

Series	Final Level	Transition	Region
Lyman	$n = 1$	$2, 3, 4, \dots \rightarrow 1$	Ultraviolet
Balmer	$n = 2$	$3, 4, 5, \dots \rightarrow 2$	Visible
Paschen	$n = 3$	$4, 5, 6, \dots \rightarrow 3$	Infrared
Brackett	$n = 4$	$5, 6, 7, \dots \rightarrow 4$	Infrared
Pfund	$n = 5$	$6, 7, 8, \dots \rightarrow 5$	Infrared

12 Rules for Electronic Configuration

The distribution of electrons in the orbitals of an atom is called **electronic configuration**. The electronic configuration of an atom is governed by the following three principles.

1. Aufbau Principle
2. Pauli's Exclusion Principle
3. Hund's Rule of Maximum Multiplicity

1. Aufbau Principle

Definition

According to the Aufbau principle, electrons occupy the orbitals in the order of increasing energy. The orbital with lower energy is filled before the orbital with higher energy.

Important Points

- Orbitals are filled in the order of increasing energy.
- The order of filling is governed by the $(n + l)$ rule.
- If two orbitals have the same value of $(n + l)$, the orbital having the lower value of n is filled first.

The order of filling electrons is

$1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p < 5s < 4d < 5p < 6s < 4f < 5d < 6p < 7s$

Example 1

Which orbital is filled first, $3d$ or $4s$?

Solution

According to the Aufbau principle,

$$4s < 3d$$

Hence,

The $4s$ orbital is filled before the $3d$ orbital.

Example 2

Which orbital is filled first, $5s$ or $4d$?

$$5s < 4d$$

Hence,

The $5s$ orbital is filled first.

2. Pauli's Exclusion Principle

Definition

According to Pauli's Exclusion Principle, no two electrons in an atom can have the same set of four quantum numbers.

Important Points

- An orbital can accommodate a maximum of two electrons.
- The two electrons must have opposite spins.
- If one electron has spin $+\frac{1}{2}$, the other must have spin $-\frac{1}{2}$.

Illustration



Allowed



Not Allowed

Example

How many electrons can occupy one orbital?

Answer

Only two electrons having opposite spins.

3. Hund's Rule of Maximum Multiplicity

Definition

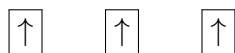
According to Hund's Rule, electrons occupy all the degenerate orbitals singly with parallel spins before pairing takes place.

Important Points

- Pairing starts only after every degenerate orbital contains one electron.
- Electrons occupying separate orbitals have parallel spins.
- This arrangement gives maximum stability.

Illustration

For the $2p^3$ configuration,



Correct filling.

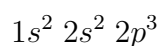
The following arrangement is incorrect.



Example

Draw the orbital diagram of nitrogen ($Z = 7$).

Electronic configuration:



Orbital representation:



This follows Hund's Rule.

Solved Examples on Electronic Configuration

Example 1

Write the electronic configuration of lithium ($Z = 3$).

Solution

Number of electrons = 3

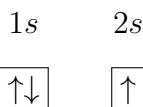
According to the Aufbau principle,

$$1s \rightarrow 2s$$

Hence,

$$1s^2 2s^1$$

Orbital diagram



Example 2

Write the electronic configuration of carbon ($Z = 6$).

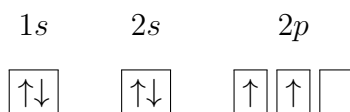
Solution

Number of electrons = 6

Electronic configuration

$$1s^2 2s^2 2p^2$$

Orbital diagram



The two electrons in the $2p$ subshell occupy different orbitals according to Hund's Rule.

Example 3

Write the electronic configuration of oxygen ($Z = 8$).

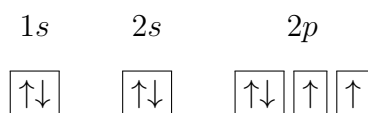
Solution

Number of electrons = 8

Electronic configuration

$$1s^2 2s^2 2p^4$$

Orbital diagram



After each $2p$ orbital receives one electron, pairing begins according to Hund's Rule.

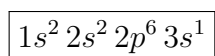
Example 4

Write the electronic configuration of sodium ($Z = 11$).

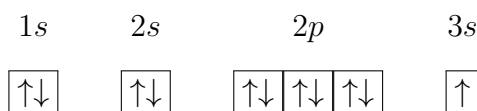
Solution

Number of electrons = 11

Electronic configuration



Orbital diagram



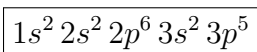
Example 5

Write the electronic configuration of chlorine ($Z = 17$).

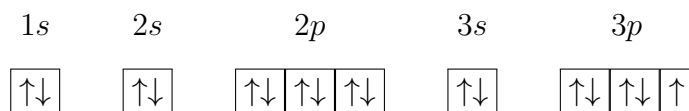
Solution

Number of electrons = 17

Electronic configuration



Orbital diagram



Practice Problems

Write the electronic configuration and orbital diagram for the following atoms.

1. Helium ($Z = 2$)
2. Beryllium ($Z = 4$)
3. Nitrogen ($Z = 7$)
4. Fluorine ($Z = 9$)

5. Neon ($Z = 10$)
6. Magnesium ($Z = 12$)
7. Aluminium ($Z = 13$)
8. Phosphorus ($Z = 15$)
9. Sulphur ($Z = 16$)
10. Argon ($Z = 18$)

Quick Revision

Rule	Statement
Aufbau Principle	Electrons occupy orbitals in the order of increasing energy.
Pauli's Exclusion Principle	An orbital can contain a maximum of two electrons with opposite spins.
Hund's Rule	Electrons occupy degenerate orbitals singly before pairing begins.

12.1 Previous Year Questions

1. What are the Postulates of Bohr's model of Hydrogen atom? Discuss the importance of this model to explain various series of Line Spectra in Hydrogen atom
2. How are the Quantum numbers n , l , m_l arrived at? Explain the significance of these Quantum numbers
3. Write the limitations of Bohr's model
4. Write the differences between Absorption and Emission Spectra